DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: B. Tech. **Branch: Common to all** **Semester: I**

Subject Code & Name: EE104 Basic Electrical Engineering

Max Marks: 60 **Date:** **Duration: 3.45 Hrs.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *Use of non-programmable scientific calculators is allowed.*
3. *Assume suitable data wherever necessary and mention it clearly.*

Marks

Q. 1 **Solve Any Two of the following.** **12**

A) Define following terms and mention their SI units: 6

- i) Electric Charge
- ii) Electric Current
- iii) Electric Potential
- iv) EMF
- v) Electric resistance
- vi) Conductance

B) How temperature affects the resistance of metals, alloys and semiconductors? What is temperature coefficient of resistance and what are its units? 6

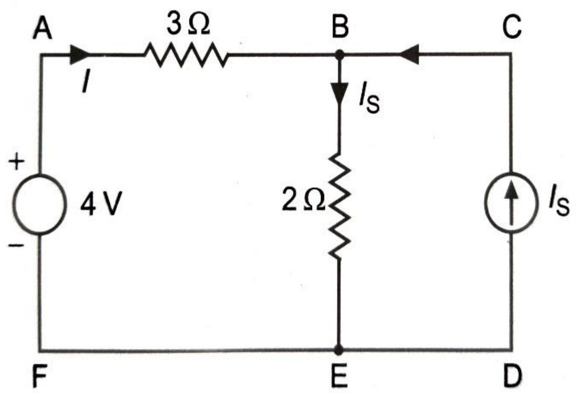
C) A resistor has a resistance of 43.6 Ω at 20° C and 50.8 Ω at 60°C. Determine the temperature coefficient of resistance of aluminum at 0 °C and at 40°C. 6

Q.2 **Solve Any Two of the following.** **12**

A) State and explain Kirchoff's current and voltage law. 6

B) State the maximum power transfer theorem. Show that for maximum power transfer $R_L = R_{th}$ and explain its importance. 6

C) In the below circuit find the value of I_s for I = 0. 6

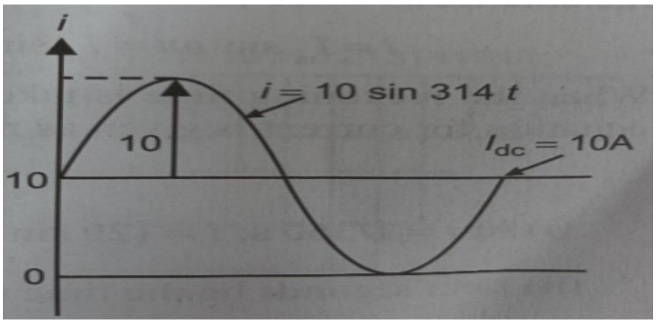


Q. 3 **Solve Any Two of the following.** **12**

A) Derive an expression for the instantaneous value of an alternating voltage varying sinusoidally. 6

B) Show that the form factor of the sinusoidal wave form is 1.11. 6

C)	Find the average and rms values of the current, I(t) = 10 + 10 sin (314t).		6



Q.4	**Solve the following.**	**12**
A)	What do you understand by real power, reactive power and apparent power? Explain with power triangle diagram.	6
B)	What do you understand by three-phase power supply? Describe star and delta connections by showing line voltages and phase voltages, line currents and phase currents.	6

Q. 5	**Solve Any Two of the following.**	**12**
A)	Draw a magnetization curve and define the hysteresis and eddy current losses.	6
B)	State and explain Flemings right-hand rule and Lenz's law.	6
C)	Differentiate between statically and dynamically induced emfs of electric and magnetic circuits.	6

***** End *****